BAHIR DAR UNIVERSITY

INSTITUTE OF TECHNOLOGY

FACULTY OF COMPUTING

DEPARTMENT OF SOFTWARE ENGINEERING
SECTION A
COURSE TITLE : OPERATING SYSTEMS AND SYSTEM PROGRAMMING
Individual Assignment on : Operating Systems

Name - Tsigereda Mulat

ID - BDU1702728

Operating System - OpenIndiana

Submitted To : Mr.Wendimu Baye

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1.Downloading and Installing the Virtual Environment Tool

To install OpenIndiana in a virtual environment, Oracle VM VirtualBox was selected as the virtualization platform. VirtualBox allows users to create and run virtual machines on a physical computer without affecting the host operating system.

The VirtualBox file was downloaded from the official Oracle VirtualBox website and installed successfully on the host computer before creating the virtual machine.

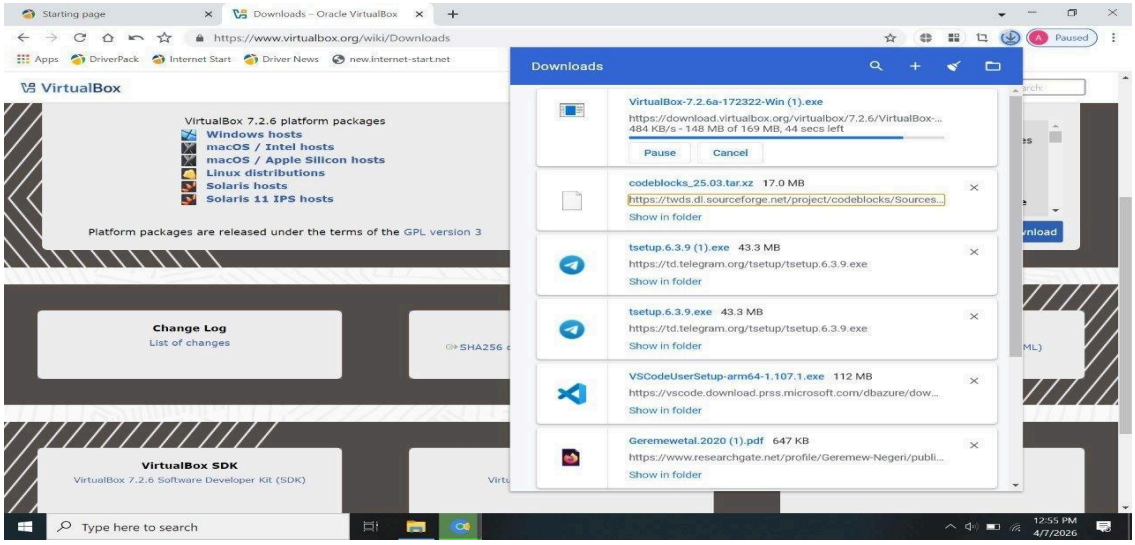

Figure 1 : Installation process of Virtual Box tool

2. Introduction

2.1 Background

OpenIndiana is an open-source Unix-based operating system derived from the OpenSolaris project. It is designed to provide a stable, secure, and enterprise-level computing environment.

OpenIndiana is widely known for supporting advanced technologies such as the ZFS file system, network virtualization, and system reliability features.
To safely test and install the operating system without affecting the main computer system, a virtual environment tool called Oracle VM VirtualBox was used. Virtualization allows multiple operating systems to run on a single physical machine by creating virtual computers known as virtual machines.

2.2 Motivation

The main motivation for this project was to gain practical experience in operating system installation and virtualization technology. This assignment also helped in understanding how virtual machines work, how operating systems are configured, and how system resources are managed in a virtual environment.
In addition, installing OpenIndiana provided hands-on experience with Unix-based operating systems and exposed advanced features such as the ZFS file system and virtual hardware configuration.

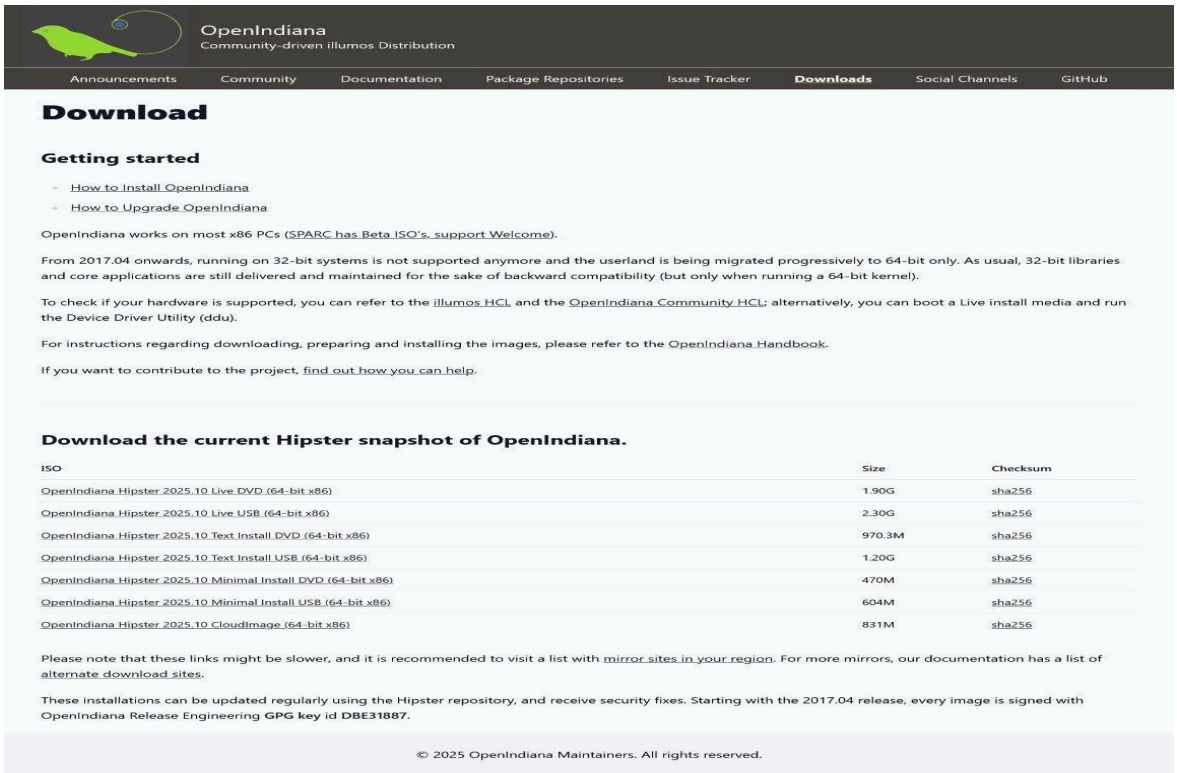

Figure 2 : Installation process of OpenINDIANA operating system

3. Objectives

The main objective of this project is to install and configure OpenIndiana in a virtualized environment using Oracle VM VirtualBox.

The specific objectives of the project are:

- To understand the concept of virtualization and virtual machines.
- To install and configure OpenIndiana successfully in a virtual environment.
- To gain practical knowledge of operating system installation procedures.
- To learn how to allocate and manage virtual hardware resources such as memory, storage, and processors.
- To study the features and filesystem support of OpenIndiana.
- To identify problems encountered during installation and provide appropriate solutions.
- To develop hands-on experience with Unix-based operating systems in a safe testing environment.

4. Requirements

Before installing OpenIndiana in a virtual environment, several hardware and software requirements were prepared to ensure that the installation process would run successfully and efficiently. Proper system requirements are important because virtualization software consumes computer resources such as memory, processor power, and storage space. Therefore, both the host machine and the virtualization software must meet the minimum requirements needed to support the operating system installation.

4.1 Hardware Requirements

Hardware resources play an important role in virtualization because the host computer must run both the main operating system and the virtual machine at the same time. Insufficient hardware

resources may lead to slow performance, installation failure, or system crashes during virtualization.

- Processor : Intel or AMD processor with virtualization technology support (Intel VT-x or AMD-V).
- Memory (RAM): Minimum of 4 GB RAM was required, although 8 GB RAM is recommended for smoother performance while running the virtual machine.
- Storage Space: At least 20 GB of free disk space was required to store the virtual machine files and the OpenIndiana installation.
- Stable Power Supply: A stable power source was necessary to avoid interruptions during installation.
- Internet Connection: Required for downloading VirtualBox and the OpenIndiana ISO installation file.

4.2 Software Requirements

Software requirements were also necessary to create and manage the virtual environment for installing OpenIndiana. The virtualization software acts as a platform that allows another operating system to run inside the host computer safely.
The following software components were required for the project:

- ➢ Oracle VM VirtualBox: Used to create and manage the virtual machine environment.
- ➢ OpenIndiana ISO File: The operating system installation image used to install OpenIndiana inside the virtual machine.
- ➢ Host Operating System: The existing operating system installed on the physical computer.
- ➢ VirtualBox Extension Support (Optional): Additional support tools for improving virtualization features and compatibility.

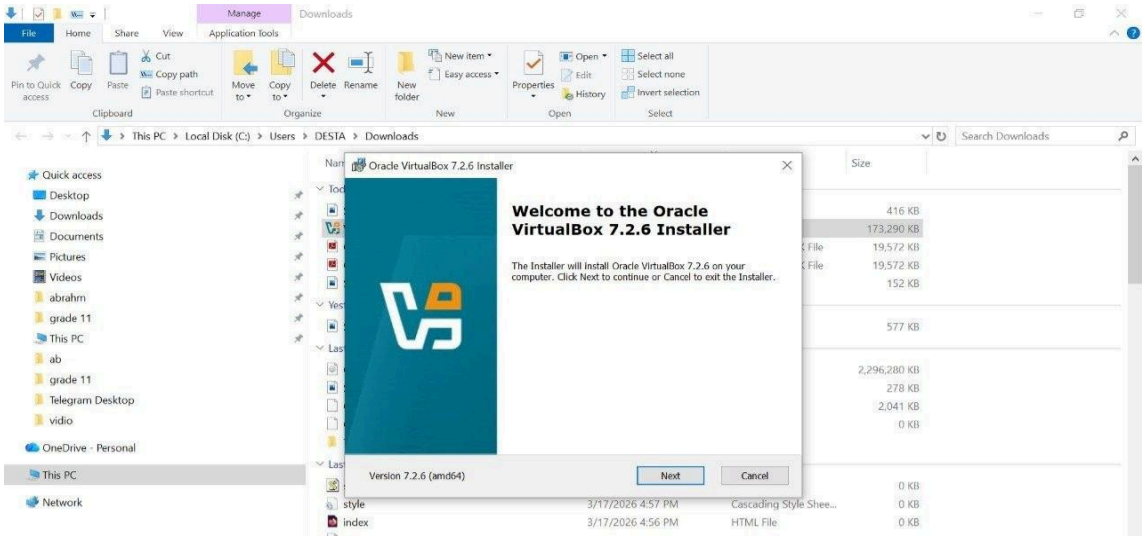

Figure 3 : Finishing installation process of Virtual Box

These software tools provided the necessary environment for installing and testing the operating system safely without modifying the main operating system of the computer.

5. Installation Steps of OpenINDIANA

The installation of OpenIndiana was performed using Oracle VM VirtualBox as the virtualization platform. The following steps were followed to successfully create the virtual machine and install the operating system.

Step 1: Launch Oracle VM VirtualBox

After installing VirtualBox, the application was opened from the computer system. The VirtualBox manager interface was used to create and manage virtual machines.

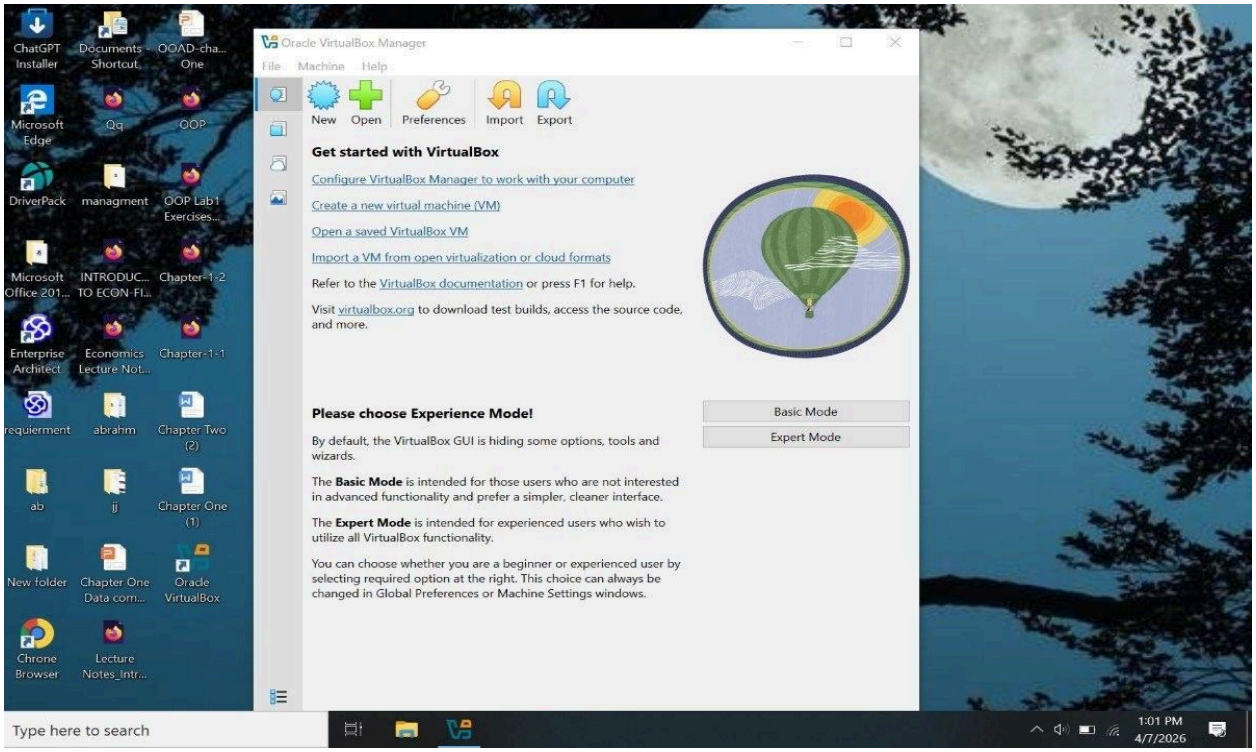

Figure 4: launching the Oracle Virtual Box

Step 2: Create a New Virtual Machine

A new virtual machine was created by clicking the New button in VirtualBox. The required information for the virtual machine was entered.

The following configurations were provided:

- Name: Use your full name as instructed by the assignment
  Example: Tsigereda Mulat
- Type: Solaris
- Version: OpenSolaris (64-bit)

This step created the basic virtual environment for the operating system installation.

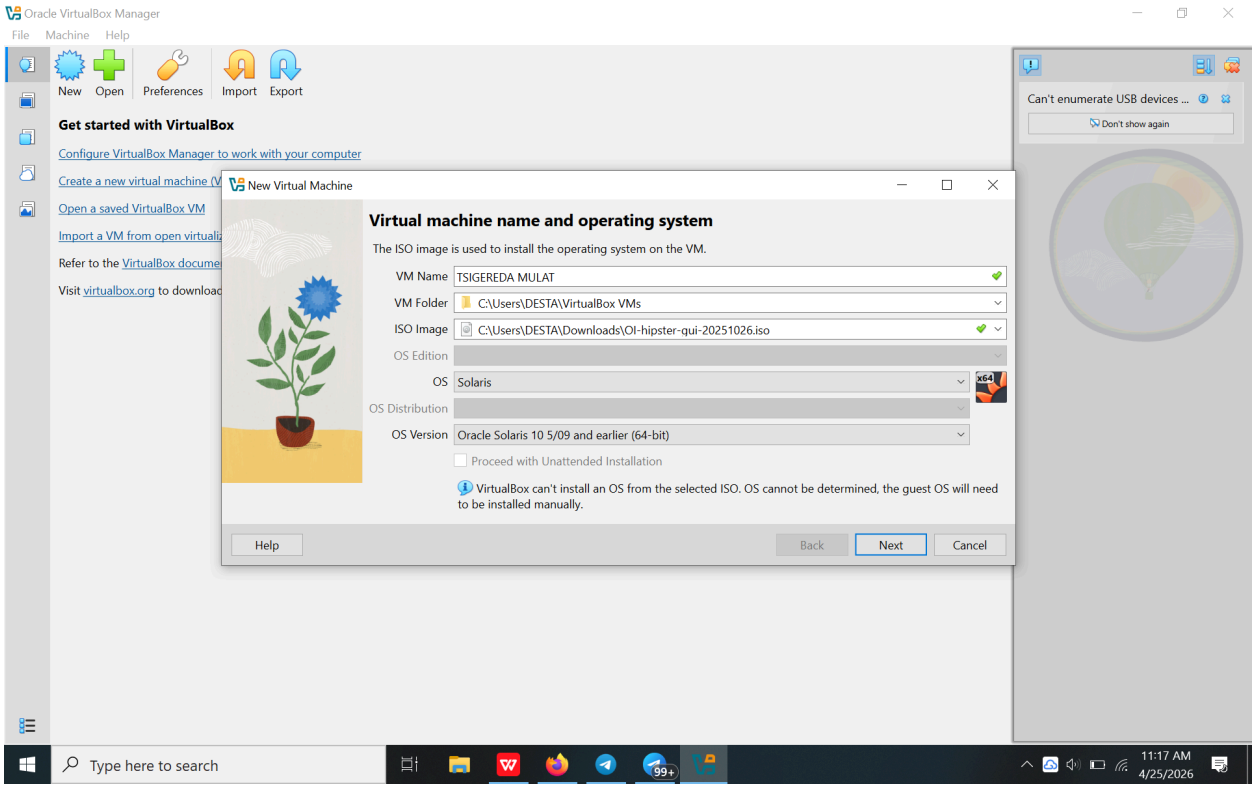

Figure 5 : Creating new Virtual Machine By using Name

Step 3: Allocate Memory (RAM)

Memory size was assigned to the virtual machine to allow the operating system to run properly. An appropriate amount of RAM was selected based on the available hardware resources of the host computer.

For this installation, sufficient memory was allocated to ensure smooth performance during installation and system operation.

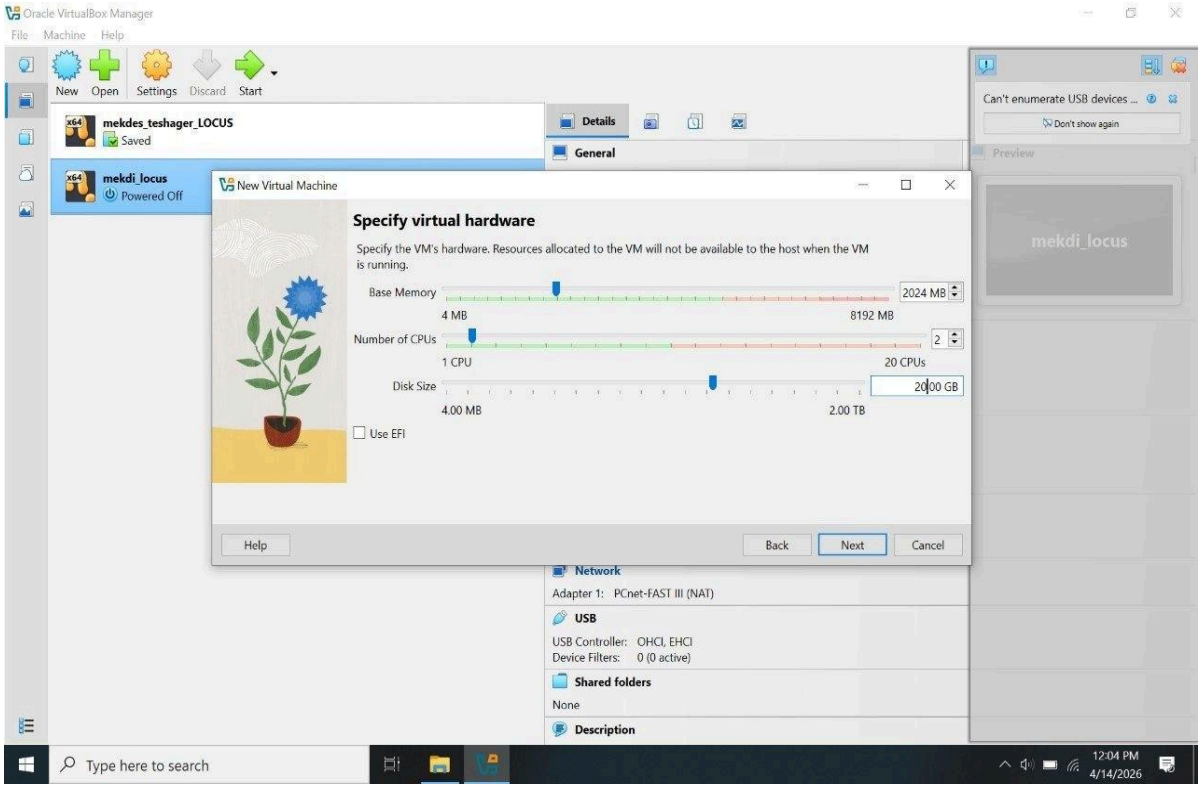

Figure 6 : Allocating memory for the New Virtual Machine

Step 4: Create a Virtual Hard Disk

A virtual hard disk was created to store the operating system files and user data inside the virtual machine.
The following settings were selected:

- Create a virtual hard disk now
- VDI (Virtual Disk Image)
- Dynamically allocated storage
- Appropriate storage size

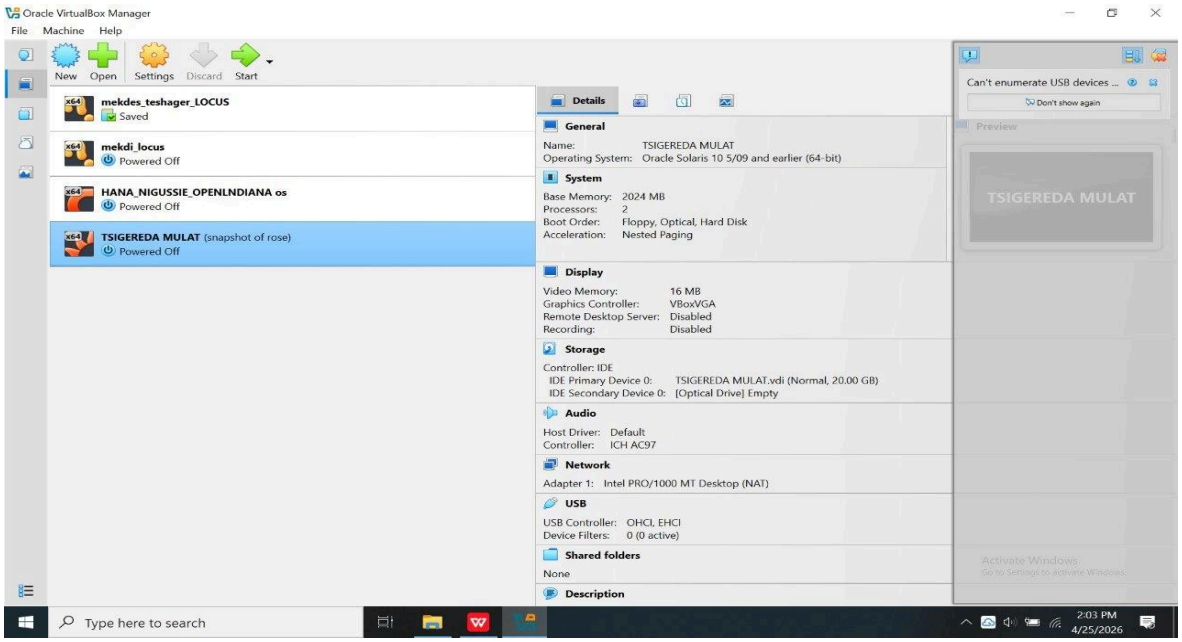

Figure 7 : Creating a Virtual Hard disk

Step 5: Attach the OpenIndiana ISO File

The downloaded OpenIndiana ISO installation file was attached to the virtual machine through the storage settings in VirtualBox. This allowed the virtual machine to boot from the installation media.

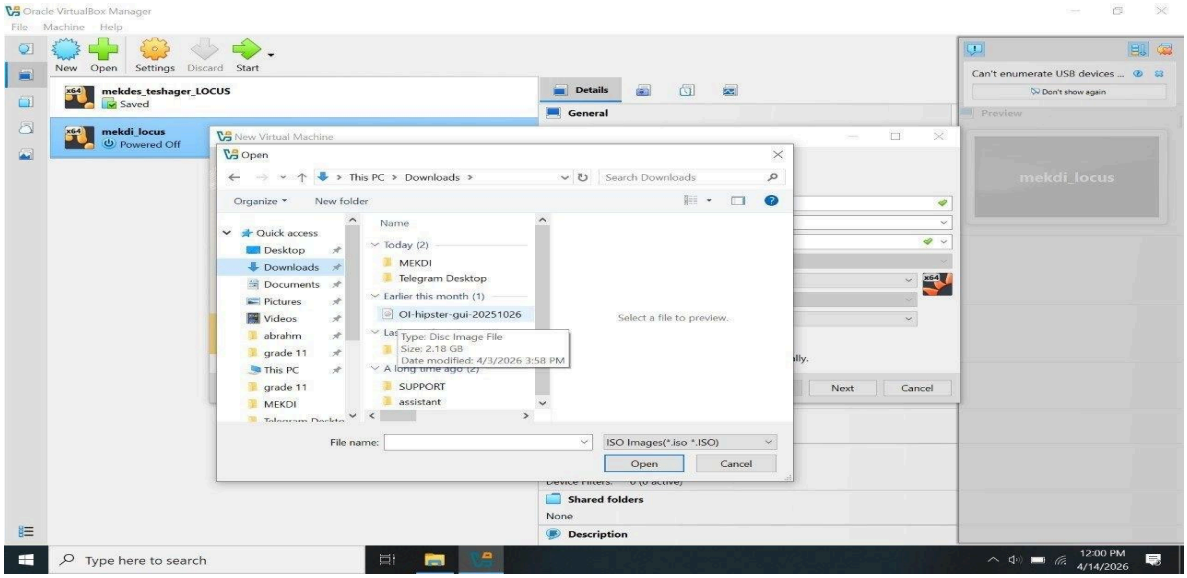

Figure 8 : Attaching the OpenIndiana file to the Virual Box

Step 6: Start the Virtual Machine

After completing the configuration, the virtual machine was started. The system booted from the OpenIndiana ISO installation file.

Figure 9 : The system start booting from OpenIndiana ISO file

Step 7: Begin OpenIndiana Installation

The installation process of OpenIndiana started after the system booted successfully. Installation options and system configurations were selected during this stage.

Figure 10 : opening the OpenIndiana operating system successfully

Step 8: Configure System Settings

During the installation of OpenIndiana, important system settings were configured to ensure proper system operation and user environment setup.
The following configurations were completed during the installation process:

- Disk configuration and storage setup
- Time zone selection
- Language and locale configuration
- Regional settings
- These settings help the operating system function correctly according to the user's geographical region and system preferences.

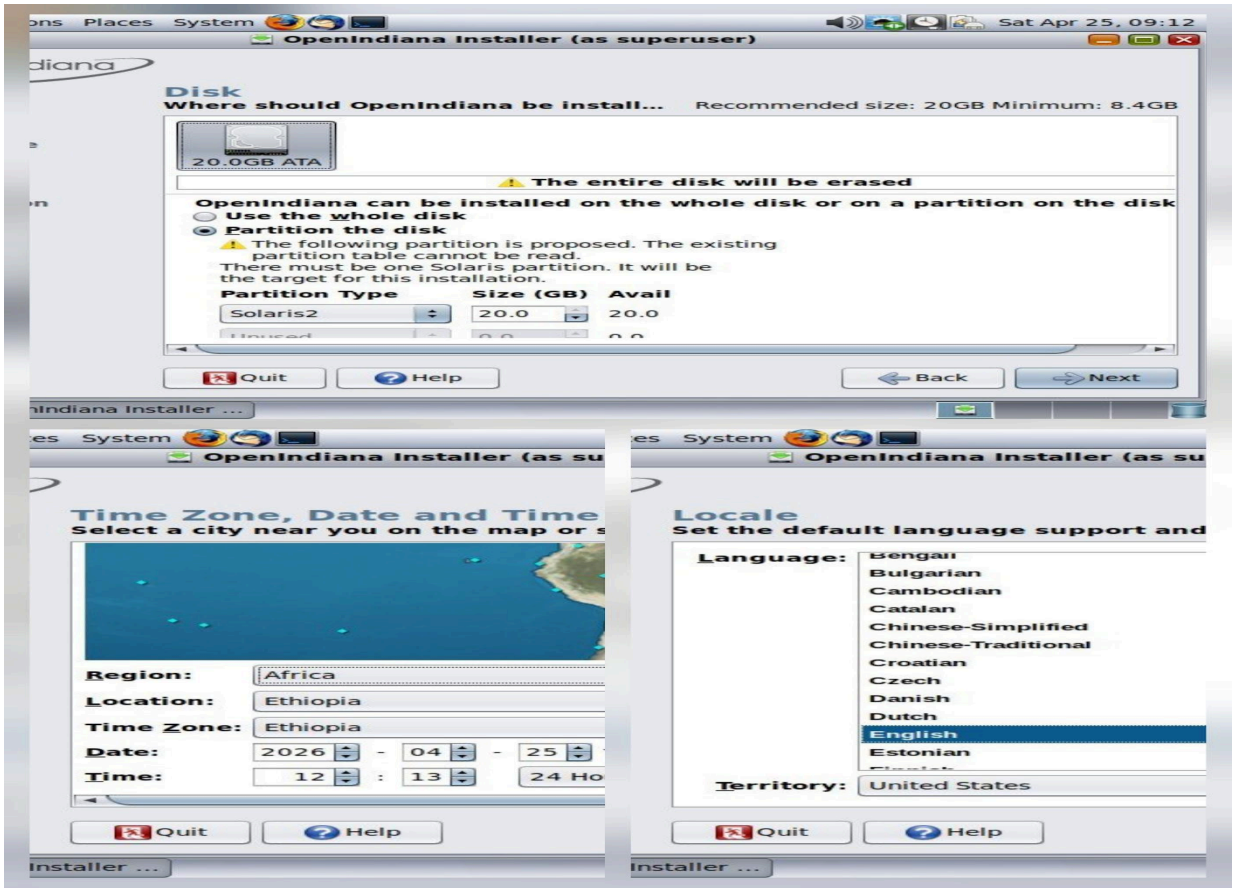

Figure 11 : Configuring system settings for OpenIndiana OS

Step 9: Configure User Account

During installation, a user account was created using the student's full name as required by the assignment instructions.The necessary account information such as username and password was configured.

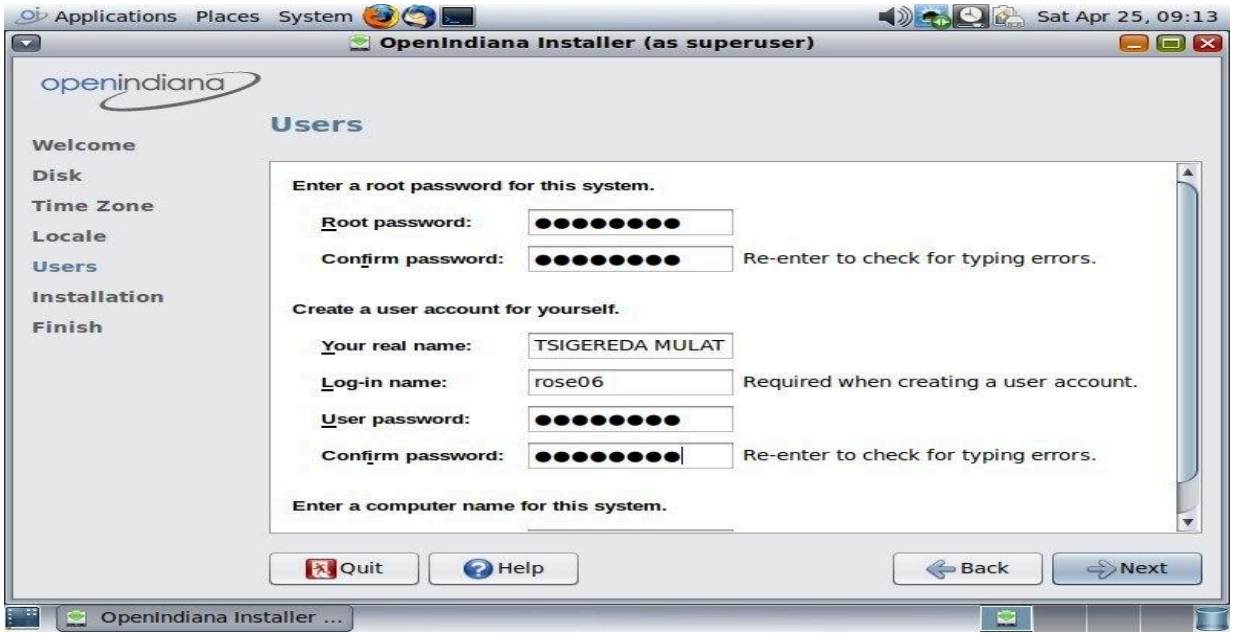

Figure 12 : configuring user Account inside the OS

Step 10: Installation Process

After completing all required configurations, the installation process of the Open INDIANA operating system started. It takes 10 -15 minutes.

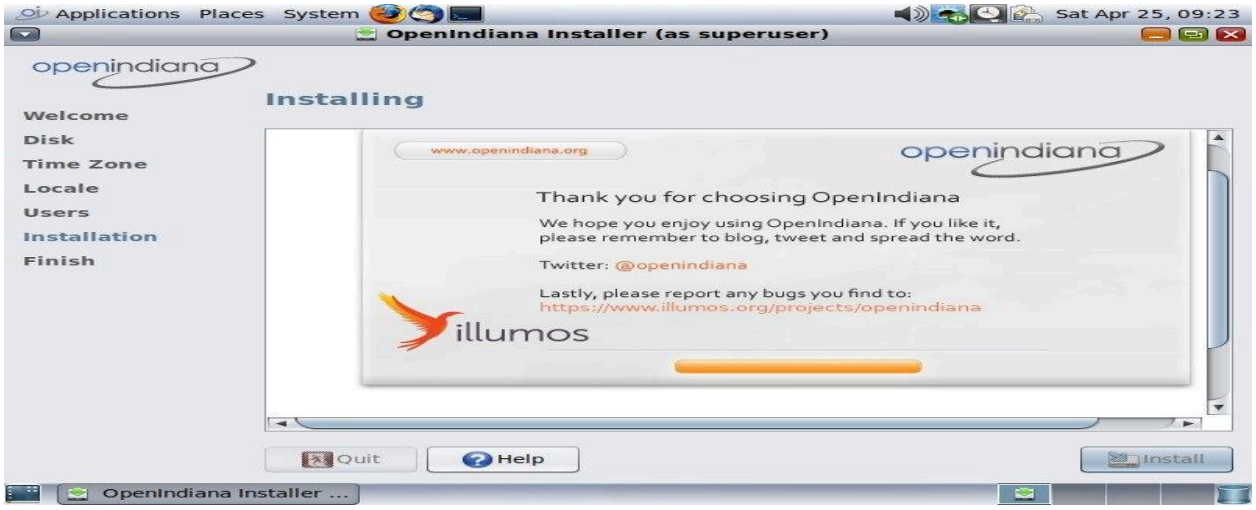

Figure 13 : Installation process of OpenIndiana Os

Step 11: Complete Installation

After completing all required configurations, the installation process finished successfully and the virtual machine restarted into the installed OpenIndiana operating system.

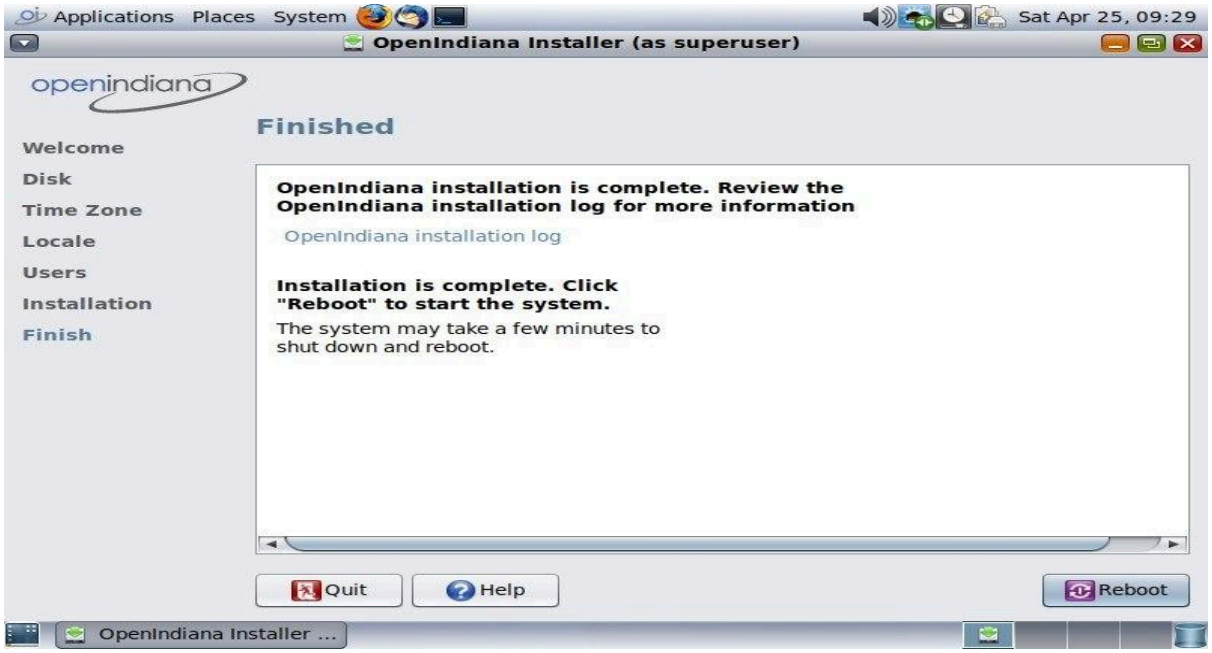

Figure 14 : Installation process of OpenIndiana is completed

Step 12. Final Result (Operating System After Installation)

After successfully completing the installation of OpenIndiana on VirtualBox, the operating system was launched without any errors. The virtual machine booted into the OpenIndiana desktop environment, confirming that the installation was successful.

The system interface appeared with a clean desktop, system menu, and basic pre-installed applications. This shows that the operating system is fully functional inside the virtual machine. At this stage, I was able to access system settings such as display, network, time zone, and user preferences. The OS responded normally, indicating that the virtual machine was correctly configured and the installation process was completed successfully.

This final result confirms that OpenIndiana is properly installed and running on VirtualBox.

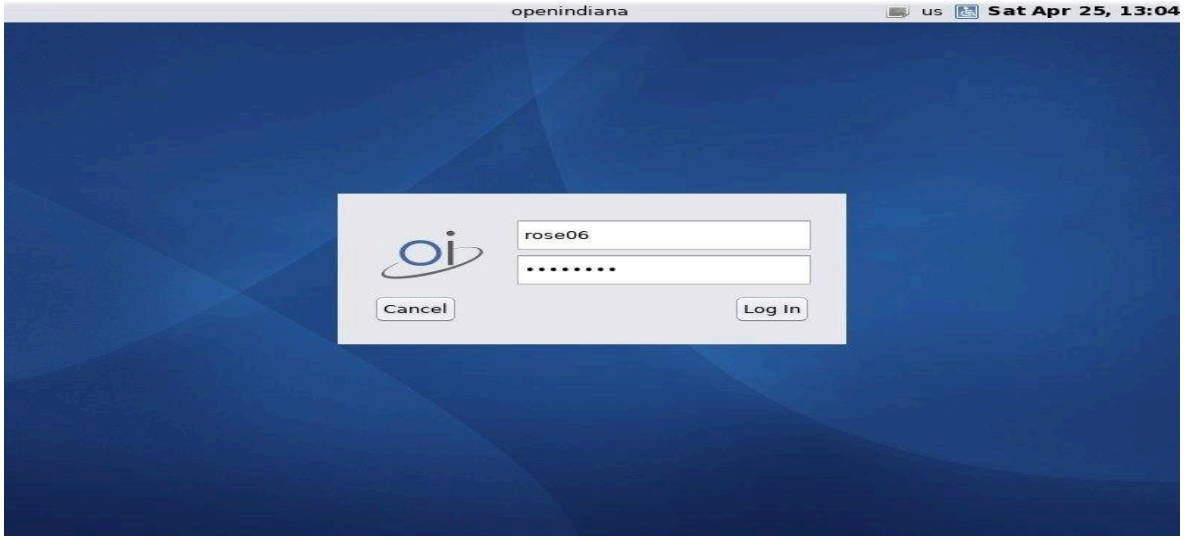

Figure 15 : Login Process of OpenIndiana Window

Figure 16 : Opened Window of OpenIndiana

6. System Call: mount (Mounts a File System)

The mount system call is used in operating systems like OpenIndiana to attach a file system to a specific directory in the system's directory tree. Once a file system is mounted, it becomes accessible to the user and applications.

In simple terms, mounting is the process of making a storage device (such as a hard disk partition, USB drive, or external storage) available for use within the operating system.

For example, when a USB drive is connected, the system uses the mount operation to link that device to a folder so that its files can be accessed.

In OpenIndiana (which is based on UNIX/Solaris systems), mount is an important system call because it allows the operating system to manage multiple storage devices and integrate them into one unified file system structure.

This system call is essential for:

- ★ Accessing external storage devices
- ★ Organizing file systems
- ★ Managing different partitions or disks

Without the mount system call, the operating system would not be able to access or use external or additional file systems properly.

Figure 17 : Opening of XTerm terminal to do system call

In Bash, the mount command is used to manually mount devices or filesystems. It helps the operating system recognize and access storage devices such as disks or partitions.

The df -h command is used to display disk space usage in a human-readable format, showing information like total size, used space, and available space for all mounted filesystems.

Together, mount shows what is connected to the system, while df -h shows how much storage is being used on those mounted filesystems.

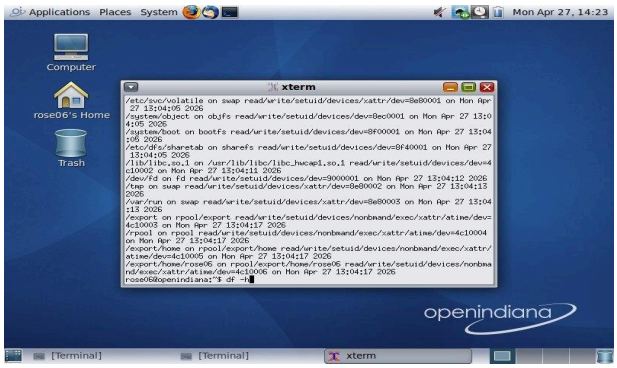

Figure 18 : Bash of mount command

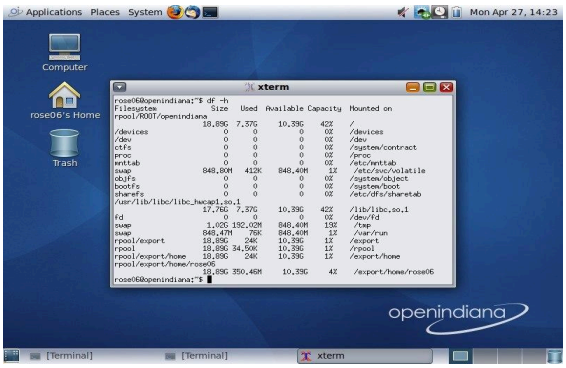

Figure 19 : Bash of df -h command

7. Issues (Problems Faced)

During the installation of OpenIndiana on VirtualBox, I faced several challenges that affected the setup process.

One of the main issues was difficulty in configuring the virtual machine settings correctly, especially selecting the appropriate operating system type and version. At first, I was not sure which option to choose for OpenIndiana, which caused confusion during setup.

Another problem was related to the ISO file not booting properly. In some cases, the virtual machine did not recognize the installation file, which required rechecking the storage settings and reattaching the ISO image.

I also experienced performance issues, such as the system running slowly due to insufficient RAM allocation. This made the installation process take longer than expected.
In addition, I faced minor errors during the installation steps, such as incorrect disk configuration or navigation mistakes in the setup interface.

At the end when I tried to do the system call on Mate Terminal which is more user friendly, I encountered an issue where I was unable to type or input commands even though the cursor was visible in the terminal window. This made it impossible to execute system commands as expected. I tested other terminals such as XTerm, which worked correctly, confirming that the problem was specific to MATE Terminal. A screenshot of the issue is attached for reference.

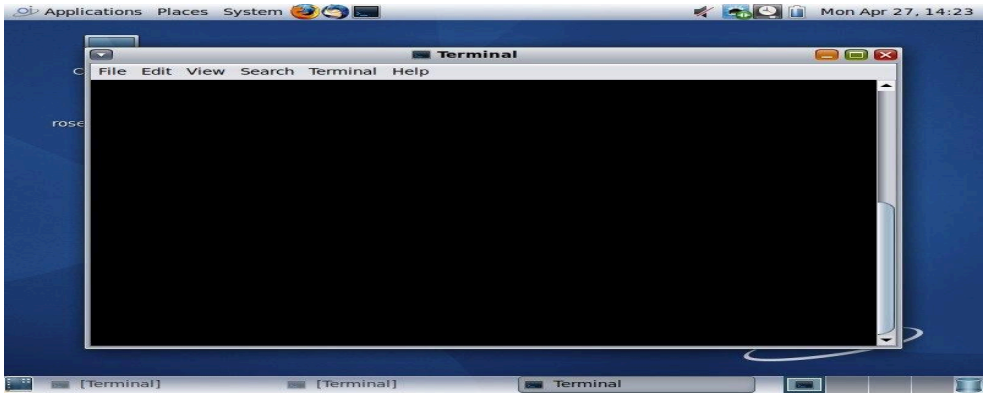

Figure 20 : The Black box screen of Mate Terminal

Despite these challenges, I was able to solve them by carefully reviewing each step, adjusting the VirtualBox settings, and restarting the installation process when necessary. This helped me successfully complete the installation.

8. Solutions to Problems Faced

To overcome the issues faced during the installation of OpenIndiana on VirtualBox, several solutions were applied.
For the problem of selecting the correct operating system type, I reviewed the VirtualBox documentation and selected the closest compatible option for OpenIndiana, which is based on Unix/Solaris systems. This helped the virtual machine to be configured correctly.

When the ISO file did not boot properly, I checked the storage settings and ensured that the OpenIndiana ISO file was correctly attached to the virtual optical drive. After correcting this, the system booted successfully.

To solve the performance issue caused by low RAM allocation, I increased the memory size in the VirtualBox settings. This improved the speed and stability of the installation process.

For installation errors related to disk setup or navigation mistakes, I restarted the installation process carefully and followed each step more attentively. This helped to avoid further errors. Overall, by adjusting the virtual machine settings and carefully repeating the installation steps, all the problems were successfully resolved.

To solve the issue with MATE Terminal not accepting input, I tried switching to another terminal (XTerm), which worked correctly. This suggested that the problem was related to the MATE Terminal configuration or session. A simple fix is to restart the terminal or log out and log back into the system. If the issue persists, resetting the MATE Terminal settings or reinstalling the MATE Terminal package can resolve the problem. After applying these steps, the terminal should allow normal command input again.

9. Filesystem Support

OpenIndiana mainly supports the ZFS (Zettabyte File System).

ZFS is the default and most important filesystem used in OpenIndiana because it is designed for high reliability, data integrity, and advanced storage management. It combines both the filesystem and volume manager in one system, which makes it very powerful.
ZFS is preferred because it has features like:

- ★ Data integrity protection (detects and prevents data corruption)
- ★ Snapshots (allows restoring previous system states easily)
- ★ High storage capacity support
- ★ Built-in RAID-like functionality for data safety

Although OpenIndiana is based on Unix/Solaris systems, it can also interact with other filesystem types through compatibility layers or external devices, such as:

- ➢ FAT32 and exFAT (for USB drives and external storage)
- ➢ NTFS (limited support, mainly for reading Windows drives)

However, filesystems like ext4, Btrfs, HFS+, and APFS are not native to OpenIndiana and are not fully supported without additional tools or workarounds.

In conclusion, OpenIndiana primarily uses ZFS because it is stable, secure, and designed for enterprise-level storage management.

10. Advantages and Disadvantages of OpenIndiana

10.1 Advantages of OpenIndiana

OpenIndiana has several advantages, especially because it is based on the Solaris operating system and uses advanced technologies like ZFS.
One of the main advantages is its high stability and reliability. It is designed to run for long periods without crashing, which makes it suitable for servers and professional environments.

Another advantage is its strong data protection using the ZFS filesystem. ZFS helps prevent data corruption and allows users to recover previous versions of files using snapshots.

OpenIndiana also has good performance for server tasks, such as networking, storage management, and system administration.

It is also free and open-source, which means users can download, modify, and use it without paying licensing fees.

In addition, it provides advanced system management tools that are useful for developers and system administrators.

10.2 Disadvantages of OpenIndiana

Although OpenIndiana has strong features, it also has some limitations.

One major disadvantage is that it has a steep learning curve. It is not very beginner-friendly, especially for users who are used to Windows or simple Linux distributions. Many system tasks require technical knowledge.

Another disadvantage is its limited software availability. Compared to Windows or popular Linux systems, OpenIndiana does not support many modern applications and games, which reduces its usability for general users.

It also has smaller community support, meaning it can be harder to find tutorials, troubleshooting guides, or quick solutions when problems occur.

In addition, hardware compatibility can sometimes be an issue. Some modern devices or drivers may not work properly or may require manual configuration.

Overall, OpenIndiana is powerful but is mainly suited for advanced users and server environments rather than everyday personal use.

11. Conclusion

In conclusion, installing OpenIndiana on VirtualBox was a valuable learning experience that helped me understand how operating systems are installed and configured in a virtual environment. Through this assignment, I learned the step-by-step process of creating a virtual machine, attaching an ISO file, and completing the installation successfully.

The project also helped me become familiar with system settings such as memory allocation, storage configuration, and boot options. In addition, I was able to explore the OpenIndiana operating system itself, including its interface and basic features after installation.

Although I faced some challenges during the process, such as configuration errors and booting issues, I was able to solve them by carefully adjusting settings and repeating certain steps. This improved my problem-solving skills and attention to detail.

Overall, this assignment strengthened my practical knowledge of operating systems and virtualization. It also showed the importance of patience and accuracy when working with system installations.

12. Future Outlook / Recommendations

In the future, OpenIndiana can be further improved and made more user-friendly by enhancing its interface and simplifying system configuration processes. This would help beginners understand and use the system more easily without requiring advanced technical knowledge
.

It is also recommended to improve software compatibility so that more modern applications can run smoothly on the system. Expanding support for popular software would increase its usability beyond server environments.

Another important improvement would be increasing community support and documentation. Having more tutorials, guides, and active forums would make it easier for users to solve problems and learn the system faster.

For users, it is recommended to use OpenIndiana mainly in a virtual environment or for learning purposes before applying it in real systems. Practicing in VirtualBox is a safe way to understand its features without affecting the main computer system.

Overall, with continuous development and better support, OpenIndiana has the potential to become a more accessible and widely used operating system in the future.

13.Virtualization in Modern Operating Systems (What, Why, and How)

13.1 What is virtualization?

Virtualization is a technology that allows a single physical computer to create and run multiple independent virtual computers, known as virtual machines. Each virtual machine behaves like a real computer with its own operating system, memory, storage, network settings, and applications. Even though they are separate environments, they all share the same physical hardware of the host system. This makes it possible to use one machine for many different computing tasks at the same time.

13.2 Why is virtualization used?

Virtualization is widely used because it improves efficiency, flexibility, and cost savings in computing systems. It allows users to run different operating systems on the same device without needing multiple computers. It is also very useful for testing and learning purposes

because users can install or experiment with software in a safe environment without damaging the main system. In addition, virtualization is important in modern IT fields such as cloud computing, data centers, and software development because it helps organizations use hardware resources more effectively and reduce maintenance costs.

13.3 How does virtualization work?

Virtualization works using a special software layer called a hypervisor, such as Oracle VirtualBox, VMware, or Hyper-V. The hypervisor sits between the physical hardware and the virtual machines. It divides the system's resources like CPU, RAM, storage, and network and allocates them to each virtual machine based on its requirements. Each virtual machine operates independently as if it is a separate computer. When a user installs an operating system like OpenIndiana on a virtual machine, the hypervisor manages all communication between the virtual OS and the actual hardware, ensuring smooth and controlled operation.